# Mathematics II

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You can find the last version of these course materials at

https://mbujosab.github.io/MatematicasII/

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## 1 Overview of **Lesson 16**

Always squared matrices in this topic

### Overview of Lesson 16

- **Eigenvalues, eigenvectors** (the prefix eigen is the German word for innate, distinct, self)
- $\left|\mathbf{A}-\lambda\mathbf{I}\right|=0$ *Characteristic equation*
- $\operatorname{tr}(\mathbf{A})$, $\det\mathbf{A}$ (demo in the next lesson)





## 2 Eigenvalues and eigenvectors

Consider the equation

$$\mathbf{A}\boldsymbol{x}=\lambda\boldsymbol{x} \qquad \text{(with} \quad \boldsymbol{x}\neq\mathbf{0}\text{)}$$

- *Eigenvalue* is any $\lambda$ such that there are solutions.
- Such *non-zero* solutions $\boldsymbol{x}$ are called *eigenvectors*.
  $\boldsymbol{x}\neq\mathbf{0}$ such that $\mathbf{A}\boldsymbol{x}$ is a multiple of $\boldsymbol{x}$

When $\lambda$ is $0$, what are the eigenvectors?

.

## 3 Example: projection matrix

- Orthogonal projection
- Which vectors are eigenvectors?
  What vectors remain in the same direction?
- What are the eigenvalues of those eigenvectors?
- Are there any other eigenvectors? With what eigenvalue?
- Two eigenspaces





## 4 Another example: swap matrix

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- A vector that does not change after the swap?
- What is the eigenvalue?
- Is there an eigenvector associated with $\lambda_2 = -1$?

$$\mathbf{A}\boldsymbol{x}_2 = -\boldsymbol{x}_2$$

Note: $\operatorname{tr}(\mathbf{A}) = 0 = \lambda_1 + \lambda_2; \qquad \det \mathbf{A} = -1 = \lambda_1 \cdot \lambda_2.$





## 5 How to find eigenvalues and eigenvectors?

How to solve

$$\mathbf{A}\boldsymbol{x} = \overbrace{\lambda}^{?}\,\overbrace{\boldsymbol{x}}^{?} \; ?$$

Let's rearrange ...

$$(\mathbf{A} - \lambda\mathbf{I})\boldsymbol{x} =$$

idea For this to hold, what kind of matrix must be $(\mathbf{A} - \lambda\mathbf{I})$?

and then what must its determinant equal? $|\mathbf{A} - \lambda\mathbf{I}| =$





## 6 How to find eigenvalues and eigenvectors?

1. Eigenvalues are the $\lambda$'s such that: $|\mathbf{A} - \lambda\mathbf{I}| =$
   ( Characteristic polynomial $P_{\mathbf{A}}(\lambda)$ )
2. How to compute $\boldsymbol{x}$ such that $(\mathbf{A} - \lambda\mathbf{I})\,\boldsymbol{x} = \mathbf{0}$?

Eigenspace (Set of eigenvectors $+\ \mathbf{0}$):

$$\mathcal{E}_\lambda(\mathbf{A}) = \left\{ \boldsymbol{x} \in \mathbb{R}^n \,\middle|\, \mathbf{A}\boldsymbol{x} = \lambda\boldsymbol{x} \right\}$$

*Spectrum*: set $\{\lambda_1, \ldots, \lambda_k\}$ of eigenvalues $\big($roots of $P_{\mathbf{A}}(\lambda)\big)$

## 7 Example (we must compute the eigenvalues first!)

**We are looking for a null determinant** (Characteristic polynomial)

$$\mathbf{A} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}; \qquad \det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix} = (3-\lambda)^2 - 1 = 0$$

Note: $\operatorname{tr}(\mathbf{A}) = 6 = \lambda_1 + \lambda_2; \qquad \det \mathbf{A} = 8 = \lambda_1 \cdot \lambda_2.$





## 8 Example (. . . and then the eigenspaces)

**And now we compute the null space** $\mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$ . . . for each $\lambda$.

For $\lambda_1 = 4$

$$(\mathbf{A} - 4\mathbf{I}) = \begin{bmatrix} 3-4 & 1 \\ 1 & 3-4 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \Rightarrow$$

For $\lambda_2 = 2$

$$(\mathbf{A} - 2\mathbf{I}) = \begin{bmatrix} 3-2 & 1 \\ 1 & 3-2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \Rightarrow$$

Are they the only two eigenvectors?

$$\mathbf{A}\boldsymbol{x}_i = \lambda \boldsymbol{x}_i; \qquad \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \boldsymbol{x}_i = \lambda \boldsymbol{x}_i.$$





## 9 Another example: 90º rotation matrix

$$\mathbf{Q} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

- What is the sum of the eigenvalues?
- What is the determinant?

Difficulties

$\lambda_1 + \lambda_2 = 0$ and $\lambda_1 \cdot \lambda_2 = 1$ $\qquad (+) \cdot (-) = (+)$?

What kind of vector can be parallel to itself after a 90º rotation?

$$\det(\mathbf{Q} - \lambda \mathbf{I}) = \begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 + 1 =$$





## 10 There are even worse examples

$$\mathbf{A} = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$$

- Eigenvalues

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 3-\lambda & 1 \\ 0 & 3-\lambda \end{vmatrix} = (3-\lambda)(3-\lambda) = 0 \begin{cases} \lambda_1 = 3 \\ \lambda_2 = 3 \end{cases}$$

- Eigenvectors
  - for $\lambda_1$: $(\mathbf{A} - \lambda \mathbf{I})\boldsymbol{x} = \mathbf{0} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \boldsymbol{x}_1; \qquad \boldsymbol{x}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$
  - for $\lambda_2$:

$\lambda = 3$ is repeated twice, but $\dim \mathcal{E}_3(\mathbf{A}) = 1$

$$\mu(3) = 2 \neq 1 = \gamma(3)$$

**Summary**:

1. The eigenvalues are those numbers $\lambda$ that make the matrix $(\mathbf{A} - \lambda\mathbf{I})$ singular. In other words, they are the roots of the characteristic polynomial: $\det(\mathbf{A} - \lambda\mathbf{I})$.
2. Any $n \times n$ matrix has a characteristic polynomial of degree $n$
3. A polynomial of degree $n$ has $n$ roots (perhaps some roots are repeated).
4. The sum of the eigenvalues of a matrix equals its trace
5. The product of the eigenvalues of a matrix equals its determinant
6. The eigenvectors associated with $\lambda$ are the non-zero vectors in $\mathcal{N}(\mathbf{A} - \lambda\mathbf{I})$.





## Questions of the Lecture 16

(L-16) Question 1. Consider the matrix

$$\mathbf{A} = \begin{bmatrix} -3 & 4 & -4 \\ -3 & 5 & -3 \\ -1 & 2 & 0 \end{bmatrix}$$

(a) The eigenvalues of $\mathbf{A}$ are $-1$, $1$, and $2$; and two corresponding eigenvectors are

$$\boldsymbol{v} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}; \qquad \boldsymbol{w} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

Verify that both vectors are indeed eigenvectors of $\mathbf{A}$. What are the corresponding eigenvalues?

(b) Find a third linearly independent eigenvector.

(L-16) Question 2. Find the eigenvalues and eigenvectors of

(a)

$$\mathbf{A} = \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$$





(b)

$$\mathbf{B} = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$$

(Strang, 2006, exercise 12 from section 5.1.)

(L-16) Question 3. If $\mathbf{B}$ has eigenvalues $1$, $2$, $3$, $\mathbf{C}$ has eigenvalues $4$, $5$, $6$, and $\mathbf{D}$ has eigenvalues $7$, $8$, $9$, what are the eigenvalues of the $6 \times 6$ matrix $\mathbf{A} = \begin{bmatrix} \mathbf{B} & \mathbf{C} \\ \mathbf{0} & \mathbf{D} \end{bmatrix}$? where $\mathbf{B}$, $\mathbf{C}$, $\mathbf{D}$ are upper triangular matrices.
(Strang, 2006, exercise 13 from section 5.1.)

(L-16) Question 4. Find the eigenvalues and eigenvectors of

(a)

$$\mathbf{A} = \begin{bmatrix} 3 & 4 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

(b)

$$\mathbf{B} = \begin{bmatrix} 0 & 0 & 2 \\ 0 & 2 & 0 \\ 2 & 0 & 0 \end{bmatrix}$$

(Strang, 2006, exercise 5 from section 5.1.)





(L-16) Question 5. The eigenvalues of $\mathbf{A}$ are equal to the eigenvalues of $\mathbf{A}^\intercal$. This is because $\det(\mathbf{A} - \lambda\mathbf{I})$ is equal to $\det(\mathbf{A}^\intercal - \lambda\mathbf{I})$.

(a) This is true because _____

(b) Show with an example that, however, the eigenvectors of $\mathbf{A}$ and $\mathbf{A}^\intercal$ are not the same.

(Strang, 2006, exercise 11 from section 5.1.)

(L-16) Question 6. Let $\mathbf{B}$ be a matrix and $\boldsymbol{x}$ an eigenvector associated with the eigenvalue $\lambda$, that is $\mathbf{B}\boldsymbol{x} = \lambda\boldsymbol{x}$; and also consider the matrix $\mathbf{A} = (\mathbf{B} + \alpha\mathbf{I})$. Prove that $\boldsymbol{x}$ is also an eigenvector of $\mathbf{A}$ with associated eigenvalue $(\lambda + \alpha)$.

(L-16) Question 7.

(a) Encuentre los autovalores y los autovectores de la matriz $\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$.

Compruebe que la traza es igual a la suma de los autovalores, y que el determinante es igual a su producto.

(b) Si consideramos una nueva matriz, generada a partir de la anterior como

$$\mathbf{B} = (\mathbf{A} - 7\mathbf{I}) = \begin{bmatrix} -6 & -1 \\ 2 & -3 \end{bmatrix}.$$

¿Cuáles son los autovalores y autovectores de la nueva matriz, y cómo están relacionados con los de $\mathbf{A}$?

(Strang, 2006, exercise 1 and 3 from section 5.1.)

(L-16) Question 8. Suppose that $\lambda$ is an eigenvalue of $\mathbf{A}$, and that $\boldsymbol{x}$ is an eigenvector such that $\mathbf{A}\boldsymbol{x} = \lambda\boldsymbol{x}$.

(a) Show that the same $\boldsymbol{x}$ is also an eigenvector of $\mathbf{B} = \mathbf{A} - 7\mathbf{I}$, and find the corresponding eigenvalue of $\mathbf{B}$.

(b) Assume that $\lambda \neq 0$ (and that $\mathbf{A}$ is invertible), show that $\boldsymbol{x}$ is also an eigenvector of $\mathbf{A}^{-1}$, and find the corresponding eigenvalue. What is its relationship with $\lambda$?

(Strang, 2006, exercise 7 from section 5.1.)

(L-16) Question 9. Suppose that $\mathbf{A}$ is an $n \times n$ matrix, such that $\mathbf{A}^2 = \mathbf{A}$. What possible values can the eigenvalues of $\mathbf{A}$ take?

(L-16) Question 10. Suppose the matrix $\underset{3\times3}{\mathbf{A}}$ has eigenvalues 1, 2, and 3. If $\boldsymbol{v}_1$ is an eigenvector associated with the eigenvalue 1, $\boldsymbol{v}_2$ is associated with the eigenvalue 2, and $\boldsymbol{v}_3$ is associated with the eigenvalue 3; then what is $\mathbf{A}(\boldsymbol{v}_1 + \boldsymbol{v}_2 - \boldsymbol{v}_3)$?

(L-16) Question 11. Provide an example that shows that eigenvalues can change when a multiple of one column is subtracted from another. Why do elimination steps not change the zero eigenvalues?
(Strang, 2006, exercise 6 from section 5.1.)





(L-16) Question 12. The characteristic polynomial of a matrix $\mathbf{A}$ can be factored as

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (\lambda_1 - \lambda)(\lambda_2 - \lambda)\cdots(\lambda_n - \lambda).$$

Show, starting from this factorization, that the determinant of $\mathbf{A}$ is equal to the product of its eigenvalues. To do this, make a smart choice of the value of $\lambda$.
(Strang, 2006, exercise 8 from section 5.1.)

(L-16) Question 13. Calculate the characteristic values (eigenvalues) and the characteristic vectors of $\mathbf{A}$ and $\mathbf{A}^2$:

$$\mathbf{A} = \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix} \quad \text{y} \quad \mathbf{A}^2 = \begin{bmatrix} 7 & -3 \\ -2 & 6 \end{bmatrix}$$

$\mathbf{A}^2$ has the same _____ as $\mathbf{A}$. When the eigenvalues of $\mathbf{A}$ are $\lambda_1$ and $\lambda_2$, the eigenvalues of $\mathbf{A}^2$ are _____.
(Strang, 2006, exercise 22 from section 5.1.)

(L-16) Question 14. Assume that the characteristic values of $\underset{3\times3}{\mathbf{A}}$ are 1, 2, and 4, what is the trace of $\mathbf{A}^2$? What is the determinant of $(\mathbf{A}^{-1})^\intercal$?
(Strang, 2006, exercise 10 from section 5.2.)





(L-16) Question 15. The equation $(\mathbf{A}^2 - 4\mathbf{I})\boldsymbol{x} = \boldsymbol{b}$ has no solution for some right-hand side $\boldsymbol{b}$. Provide as precise information as possible about the eigenvalues of the matrix $\mathbf{A}$ (the matrix $\mathbf{A}$ is diagonalizable).

(L-16) Question 16. You are given the matrix

$$\mathbf{A} = \begin{bmatrix} 0.5 & 0.2 & 0.2 \\ 0.1 & 0.5 & 0.5 \\ 0.4 & 0.3 & 0.3 \end{bmatrix}$$

One of the eigenvalues is $\lambda = 1$. What are the eigenvalues of $\mathbf{A}$? [Hint: Very little calculation is required! You should be able to see another eigenvalue by inspecting the form of $\mathbf{A}$, and the third through a simple calculation. You should not need to calculate $\det(\mathbf{A} - \lambda\mathbf{I})$ unless you really want to do it the complicated way.]





## 1 Highlights of Lesson 17

### Highlights of Lesson 17

- Similar matrices: $\mathbf{C} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}$
- Triangular block diagonalization of a matrix

$$\begin{bmatrix} \mathbf{A} \\ \mathbf{I} \end{bmatrix} \xrightarrow[esp(\boldsymbol{\tau}_p^{-1}\ldots\boldsymbol{\tau}_1^{-1})]{\boldsymbol{\tau}_1\ldots\boldsymbol{\tau}_p} \begin{bmatrix} \mathbf{C} \\ \mathbf{S} \end{bmatrix} \qquad \text{where} \quad \mathbf{S} = \mathbf{I}_{\boldsymbol{\tau}_1\ldots\boldsymbol{\tau}_p}.$$

- Diagonalizable matrices: when $\mathbf{C}$ is diagonal.

## 2 Similar matrices

**A** and **C** are *similar* if there exists an invertible **S** such that

$$\mathbf{C} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}$$

If **A** and **C** are similar (*see demos in the book*):

- The same determinant: $\det \mathbf{A} = \det \mathbf{C}$
- The same characteristic polynomial: $|\mathbf{A} - \lambda\mathbf{I}| = |\mathbf{C} - \lambda\mathbf{I}|$
- The same eigenvalues (with the same *algebraic* and *geometric* multiplicities).
- The same trace: $\left(\mathbf{A}\mathbf{I}_{\underset{[(\alpha)i+j]}{\tau}}\right)_{|j} = \alpha(\mathbf{A}_{|i}) + \mathbf{A}_{|j} \quad \Rightarrow$

$$_{j|}\left(\mathbf{A}\mathbf{I}_{\underset{[(\alpha)i+j]}{\tau}}\right)_{|j} = \alpha({}_{j|}\mathbf{A}_{|i}) + {}_{j|}\mathbf{A}_{|j}$$

*Mirror* inverse transf.: $\left(\mathbf{I}_{(\tau_1\cdots\tau_k)}\right)^{-1} = {}_{esp(\tau_k^{-1}\cdots\tau_1^{-1})}\mathbf{I}$

$$\mathbf{I} = \mathbf{I}_{\underset{[(-\alpha)j+i]}{\tau}}\mathbf{I}_{\underset{[(\alpha)i+j]}{\tau}} = \mathbf{I}_{\underset{[(\frac{1}{\alpha})j]}{\tau}}\mathbf{I}_{\underset{[(\alpha)j]}{\tau}} \quad \Rightarrow \quad \mathbf{A} \text{ similar to } {}_{esp(\tau_1\cdots\tau_k)^{-1}}\mathbf{A}_{\tau_1\cdots\tau_k}$$





## 3 Block diagonalization of a matrix (toothed matrix)

Consider $\mathbf{A} = \left[\begin{array}{c|c} \mathbf{C} & \\ \hline * & \mathbf{L} \end{array}\right] \in \mathbb{C}^{n\times n}$ where

**C** (of order $m$) is singular and **L** is full rank lower triangular; then there exists an invertible **R** such that

$$\mathbf{R}^{-1}\mathbf{A}\mathbf{R} = \left[\begin{array}{c|c|c} & 0 & \\ * & \vdots & \\ {\scriptstyle m\times(m-1)} & 0 & \\ \hline & d_{m+1} & \beta_{m+1} \\ & d_{m+2} & * \quad \beta_{m+2} \\ * & \vdots & * \quad * \quad \ddots \\ & d_n & * \quad * \quad \cdots \quad \beta_n \end{array}\right]$$

$$\left(\ldots_{\underset{[(-\alpha_j)m+j]}{\tau}}\ldots\right)\mathbf{A}\left(\ldots_{\underset{[(\alpha_j)j+m]}{\tau}}\ldots\right); \qquad j = 1,\ldots,m-1.$$





## 4 Block diagonalization of a matrix (toothed matrix)

Consider $\mathbf{A} = \left[\begin{array}{c|c} \mathbf{C} & \\ \hline * & \mathbf{L} \end{array}\right] \in \mathbb{C}^{n\times n}$ where

**C** (of order $m$) is singular and **L** is full rank lower triangular, then there exists $\mathbf{S} = \mathbf{R}\mathbf{P}$ (invertible) such that

$$\mathbf{P}^{-1}\mathbf{R}^{-1}\mathbf{A}\mathbf{R}\mathbf{P} = \left[\begin{array}{c|c|c} & 0 & \\ * & \vdots & \\ {\scriptstyle m\times(m-1)} & 0 & \\ \hline & 0 & \beta_{m+1} \\ & 0 & * \quad \beta_{m+2} \\ * & \vdots & * \quad * \quad \ddots \\ & 0 & * \quad * \quad \cdots \quad \beta_n \end{array}\right]$$

$$\left(\ldots_{\underset{[(-\alpha_j)m+j]}{\tau}}\ldots\right)\mathbf{R}^{-1}\mathbf{A}\mathbf{R}\left(\ldots_{\underset{[(\alpha_j)j+m]}{\tau}}\ldots\right); \qquad j = m+1,\ldots,n.$$





## 5 A very simple example

**Example**

Consider $\mathbf{A} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & -2 & 1 \end{bmatrix}$ with eigenvalues 0, 1 and 1.

$$\left[\begin{array}{c}\mathbf{A}\\ \hline \mathbf{I}\end{array}\right] \xrightarrow{(-)} \left[\begin{array}{c}0\\ \hline \mathbf{I}\end{array}\right] \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & -2 & 1 \\ \hline 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow[\substack{[(1)\mathbf{1}+\mathbf{2}]\\ [(2)\mathbf{3}+\mathbf{2}]\\ [\mathbf{2}\rightleftharpoons\mathbf{3}]}]{\tau} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow[\substack{[\mathbf{2}\rightleftharpoons\mathbf{3}]\\ [(-2)\mathbf{2}+\mathbf{3}]\\ [(-1)\mathbf{2}+\mathbf{1}]}]{\tau} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ \hline 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow{(+)} \left[\begin{array}{c}\mathbf{C}\\ \hline \mathbf{S}\end{array}\right]$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{\text{diagonal}}$$

## 6 A not so simple example

**Example**

Consider $\mathbf{A} = \begin{bmatrix} -2 & 0 & 3 \\ 3 & -2 & -9 \\ -1 & 2 & 6 \end{bmatrix}$ with eigenvalues 1, 1 and 0.

$$\begin{bmatrix} -3 & 0 & 3 \\ 3 & -3 & -9 \\ -1 & 2 & 5 \end{bmatrix} \xrightarrow{(-)\,1\mathbf{I}} \left[\begin{array}{ccc} -3 & 0 & 3 \\ 3 & -3 & -9 \\ -1 & 2 & 5 \\ \hline 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[[(-2)\mathbf{2}+\mathbf{3}]]{\tau\,[(1)\mathbf{1}+\mathbf{3}]} \left[\begin{array}{ccc} -3 & 0 & 0 \\ 3 & -3 & 0 \\ -1 & 2 & 0 \\ \hline 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[[(-1)\mathbf{3}+\mathbf{1}]]{\tau\,[(2)\mathbf{3}+\mathbf{2}]} \left[\begin{array}{ccc} -2 & -2 & 0 \\ 1 & 1 & 0 \\ -1 & 2 & 0 \\ \hline 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow{(+)\,1\mathbf{I}}$$

$$\xrightarrow{(-)\,1\mathbf{I}} \left[\begin{array}{ccc} -2 & -2 & 0 \\ 1 & 1 & 0 \\ \hline -1 & 2 & 0 \\ \hline 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow{\tau\,[(-1)\mathbf{1}+\mathbf{2}]} \left[\begin{array}{ccc} -2 & 0 & 0 \\ 1 & 0 & 0 \\ \hline -1 & 3 & 0 \\ \hline 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[[(1)\mathbf{2}+\mathbf{1}]]{\tau} \left[\begin{array}{ccc} -1 & 0 & 0 \\ 1 & 0 & 0 \\ \hline -1 & 3 & 0 \\ \hline 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow{(+)\,1\mathbf{I}}$$

$$\xrightarrow{(-)\,0\mathbf{I}} \left[\begin{array}{ccc} 0 & 0 & 0 \\ \hline 1 & 1 & 0 \\ -1 & 3 & 1 \\ \hline 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[[(4)\mathbf{3}+\mathbf{1}]]{\tau\,[(-1)\mathbf{2}+\mathbf{1}]} \left[\begin{array}{ccc} 0 & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & 3 & 1 \\ \hline 6 & -1 & 1 \\ -9 & 1 & -2 \\ 4 & 0 & 1 \end{array}\right] \xrightarrow[[(1)\mathbf{1}+\mathbf{2}]]{\tau\,[(-4)\mathbf{1}+\mathbf{3}]} \left[\begin{array}{ccc} 0 & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & 3 & 1 \\ \hline 6 & -1 & 1 \\ -9 & 1 & -2 \\ 4 & 0 & 1 \end{array}\right] \xrightarrow{(+)\,0\mathbf{I}}$$





## 7 Every matrix is similar to a toothed matrix

For every $\mathbf{A}$ there exists $\mathbf{S}$ such that

$$\mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \mathbf{C} \quad \Rightarrow \quad \mathbf{A}\mathbf{S} = \mathbf{S}\mathbf{C}$$

where $\mathbf{C}$, toothed, has the eigenvalues on the diagonal

**Example**

$$\begin{bmatrix} 6 & -1 & 1 \\ -9 & 1 & -2 \\ 4 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} -2 & 0 & 3 \\ 3 & -2 & -9 \\ -1 & 2 & 6 \end{bmatrix} \begin{bmatrix} 6 & -1 & 1 \\ -9 & 1 & -2 \\ 4 & 0 & 1 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}}_{\text{toothed}}$$

Consequences

- $\sum \lambda_i = \operatorname{tr}(\mathbf{A})$ and $\prod \lambda_i = \det \mathbf{A}$
- $\mathbf{A}\mathbf{S}_{|j} = \mathbf{S}\mathbf{C}_{|j} \quad \Rightarrow$ for $j$ such that $\mathbf{C}_{|j} = \lambda_i \mathbf{I}_{|j}$ :

$$\mathbf{A}(\mathbf{S}_{|j}) = \mathbf{S}(\lambda_i \mathbf{I}_{|j}) = \lambda_i(\mathbf{S}_{|j}) \quad \Rightarrow \quad \mathbf{S}_{|j} \text{ is an eigenvector.}$$





## 8 Back to the simple, "toothless" example

Consider $\mathbf{A} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & -2 & 1 \end{bmatrix}$ with eigenvalues 0, 1 and 1.

$$\begin{bmatrix} \mathbf{A} \\ \hline \mathbf{I} \end{bmatrix} \xrightarrow{(-)\,0\mathbf{I}} \left[\begin{array}{ccc} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & -2 & 1 \\ \hline 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[[2 \rightleftharpoons 3]]{\tau\,[(1)\mathbf{1}+\mathbf{2}]\,[(2)\mathbf{3}+\mathbf{2}]} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right] \xrightarrow[[(-1)\mathbf{2}+\mathbf{1}]]{\tau\,[2 \rightleftharpoons 3]\,[(-2)\mathbf{2}+\mathbf{3}]} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ \hline 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right] \xrightarrow{(+)\,0\mathbf{I}} \begin{bmatrix} \mathbf{C} \\ \hline \mathbf{S} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{A}(\mathbf{S}_{|j}) = \lambda_i(\mathbf{S}_{|j}) \quad \Rightarrow \quad \mathbf{S}_{|j} \text{ is an eigenvector.}$$





## 9 Diagonalizable matrices

- A matrix is diagonalizable **if and only if** *algebraic* and *geometric* multiplicities are equal for each eigenvalue
- If there are no repeated eigenvalues, there are no "teeth" either
- When there are no repeated eigenvalues $\underset{n \times n}{\mathbf{A}}$ is diagonalizable (is guaranteed to have $n$ independent eigenvectors)

## 10 Diagonalizing a matrix

- Find the spectrum: $\{\lambda_1, \lambda_2, \ldots\}$
- Find the *algebraic multiplicity* of each eigenvalue: $\mu(\lambda_i)$

then choose one of these alternatives:

1. teething the matrix (implemented in NAcAL)
2. . . . or for each $\lambda_i$
   - find the eigenspace

$$\mathcal{E}_{\lambda_i}(\mathbf{A}) = \left\{ \boldsymbol{x} \in \mathbb{R}^n \,\middle|\, \mathbf{A}\boldsymbol{x} = \lambda_i \boldsymbol{x} \right\} = \mathcal{N}(\mathbf{A} - \lambda_i \mathbf{I}).$$

   - check $\mu(\lambda_i) = \dim \mathcal{E}_{\lambda_i}(\mathbf{A})$ (algebraic and geometric multiplicities are equal)

$$\mathbf{D} = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_k \end{bmatrix}; \quad \mathbf{S} = \Big[ \text{Basis for } \mathcal{E}_{\lambda_1}(\mathbf{A}) \# \cdots \# \text{Basis for } \mathcal{E}_{\lambda_k}(\mathbf{A}) \Big]$$

$$\mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \mathbf{D} \quad \Leftrightarrow \quad \mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$$





## 11 Powers of a Diagonalizable Matrix

If $\mathbf{A}\boldsymbol{x} = \lambda\boldsymbol{x}$ then $\mathbf{A}^2\boldsymbol{x} = \mathbf{A}\mathbf{A}\boldsymbol{x} = \mathbf{A}(\lambda\boldsymbol{x}) = \lambda\mathbf{A}\boldsymbol{x} =$

- What is the relationship between the eigenvectors of $\mathbf{A}$ and those of $\mathbf{A}^2$?

- What is the relationship between the eigenvalues of $\mathbf{A}$ and those of $\mathbf{A}^2$?

If $\mathbf{A}$ is diagonalizable and invertible, we have that

$$\mathbf{A}^{-1} = \big(\mathbf{S}\mathbf{D}(\mathbf{S}^{-1})\big)^{-1} = \mathbf{S}\mathbf{D}^{-1}(\mathbf{S}^{-1}).$$

Moreover, if we follow the following convention

$$\mathbf{A}^0 = \mathbf{A}^{(1-1)} = \mathbf{A}^1(\mathbf{A}^{-1}) = \mathbf{I};$$

then, for every diagonalizable and invertible $\mathbf{A}$ and every integer $n$:
$\mathbf{A}^n = \mathbf{S}\mathbf{D}(\mathbf{S}^{-1}) \cdot \cdots \cdot \mathbf{S}\mathbf{D}(\mathbf{S}^{-1}) = \mathbf{S}\mathbf{D}^n(\mathbf{S}^{-1})$.





## Questions of the Lecture 17

(L-17) Question 1. Factor these matrices into $\mathbf{S}\mathbf{D}\mathbf{S}^{-1}$;

(a) $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$

(b) $\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$

(Strang, 2006, exercise 15 from section 5.2.)

(L-17) Question 2. Which of these matrices cannot be diagonalized?

(a)
$$\mathbf{A}_1 = \begin{bmatrix} 2 & -2 \\ 2 & -2 \end{bmatrix}$$

(b)
$$\mathbf{A}_2 = \begin{bmatrix} 2 & 0 \\ 2 & -2 \end{bmatrix}$$

(c)
$$\mathbf{A}_3 = \begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix}$$

(Strang, 2006, exercise 5 from section 5.2.)





(L-17) Question 3. If $\mathbf{A} = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$ find $\mathbf{A}^{100}$ by diagonalizing $\mathbf{A}$.

(Strang, 2006, exercise 7 from section 5.2.)

(L-17) Question 4. If the eigenvalues of $\underset{3\times 3}{\mathbf{A}}$ are 1, 1, and 2, which of the following statements can we affirm to be true?

(a) $\mathbf{A}$ is invertible.
(b) $\mathbf{A}$ is diagonalizable.
(c) $\mathbf{A}$ is not diagonalizable.

(Strang, 2006, exercise 11 from section 5.2.)

(L-17) Question 5. Let $\mathbf{A}$ be the matrix

$$\mathbf{A} = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 2 \end{bmatrix}$$

(a) ($1^{\text{pts}}$) Determine if $\mathbf{A}$ is diagonalizable, and if so, find a diagonal matrix $\mathbf{D}$ and a matrix $\mathbf{S}$ such that $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$.
(b) ($0.5^{\text{pts}}$) Compute $\left(\mathbf{A}^6\right)\boldsymbol{v}$, where $\boldsymbol{v} = (0, \quad 0, \quad 0, \quad 1,)$.
(c) ($0.5^{\text{pts}}$) Using the values obtained in the first part, justify that $\mathbf{A}$ is invertible.
(d) ($0.5^{\text{pts}}$) What is the relationship between the eigenvalues of $\mathbf{A}$ and those of $\mathbf{A}^{-1}$?

(L-17) Question 6. If $\mathbf{A} = \mathbf{SDS}^{-1}$; then $\mathbf{A}^3 = (\quad)(\quad)(\quad)$ and $\mathbf{A}^{-1} = (\quad)(\quad)(\quad)$.
(Strang, 2006, exercise 16 from section 5.2.)

(L-17) Question 7. Consider the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}$$

(a) Find the eigenvalues of $\mathbf{A}$.
(b) Find the eigenvectors of $\mathbf{A}$.
(c) Diagonalize $\mathbf{A}$: write it as $\mathbf{A} = \mathbf{SDS}^{-1}$.

(L-17) Question 8. False or true? If the eigenvalues of $\mathbf{A}$ are 2, 2, and 3, then we know that the matrix is
(a) Invertible
(b) Diagonalizable
(c) Not diagonalizable.

(L-17) Question 9. Let the matrices

$$\mathbf{A}_1 = \begin{bmatrix} 8 & \\ & 2 \end{bmatrix}; \qquad \mathbf{A}_2 = \begin{bmatrix} 9 & 4 \\ & 1 \end{bmatrix}; \qquad \mathbf{A}_3 = \begin{bmatrix} 10 & 5 \\ -5 & \end{bmatrix}$$





(a) Complete these matrices so that in all three cases $\det \mathbf{A}_i = 25$. Thus, the trace is equal to 10 in all cases, and therefore for the three matrices, the only eigenvalue $\lambda = 5$ is repeated twice ($\lambda^2 = 25$ and $\lambda + \lambda = 10$ imply $\lambda = 5$).
(b) Find a characteristic vector with $\mathbf{A}\boldsymbol{x} = 5\boldsymbol{x}$. These three matrices are not diagonalizable because there is not a second linearly independent eigenvector from the first.

(Strang, 2006, exercise 27 from section 5.2.)

(L-17) Question 10. Factor the following matrices into $\mathbf{S}\ \mathbf{D}\ \mathbf{S}^{-1}$.

(a) $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$
(b) $\mathbf{B} = \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$

(Strang, 2006, exercise 1 from section 5.2.)

(L-17) Question 11. Find the matrix $\mathbf{A}$ whose eigenvalues are 1 and 4, whose eigenvectors are $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ respectively.
(Strang, 2006, exercise 2 from section 5.2.)

(L-17) Question 12. If the diagonal elements of an upper triangular matrix of order $3 \times 3$ are 1, 2, and 7, can you determine if the matrix is diagonalizable? What is $\mathbf{D}$?
(Strang, 2006, exercise 4 from section 5.2.)

(L-17) Question 13.





(a) Find the eigenvalues and eigenvectors of the matrix $\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$.
(b) Explain why (or why not) the matrix $\mathbf{A}$ is diagonalizable.

(L-17) Question 14. Let $\mathbf{A}$ be a $3 \times 3$ matrix. Assume that its eigenvalues are 1 and 0, that a basis of the eigenvectors associated with $\lambda = 1$ are $[1, 0, 1]$ and $[0, 0, 1]$; while those associated with $\lambda = 0$ are parallel to $[1, 1, 2]$.
(a) Is $\mathbf{A}$ diagonalizable? If so, write the diagonal matrix $\mathbf{D}$ and the matrix $\mathbf{S}$ such that $\mathbf{A} = \mathbf{SDS}^{-1}$.
(b) Find $\mathbf{A}$.

(L-17) Question 15. Let $\mathbf{A}$ be a $2 \times 2$ matrix such that $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ is an eigenvector for $\mathbf{A}$ with eigenvalue 2, and $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ is another eigenvector for $\mathbf{A}$ with eigenvalue -2. If $\boldsymbol{v} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, compute $\left(\mathbf{A}^3\right)\boldsymbol{v}$.





## 1 Highlights of Lesson 18

### Highlights of Lesson 18

- Symmetric matrices $\mathbf{A} = \mathbf{A}^{\mathsf{T}}$
  - Eigenvalues and eigenvectors
- Introd. positive definite matrices

## 2 Symmetric Matrices $\mathbf{A} = \mathbf{A}^\mathsf{T}$

What is special about $\mathbf{A}\boldsymbol{x} = \lambda\boldsymbol{x}$ when $\underset{n\times n}{\mathbf{A}}$ is symmetric?

1. The eigenvalues are REAL
2. n eigenvectors *can be chosen* ORTHOGONAL

(Always diagonalizable)

The usual diagonalizable case:

$$\mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \mathbf{D} \quad \longleftrightarrow \quad \mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$$

Symmetric case:

I can choose orthonormal eigenvectors (ortho*normal* columns of $\mathbf{S} = \mathbf{Q}$)

(If $\mathbf{A} = \mathbf{A}^\mathsf{T}$) $\quad \mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{-1} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\mathsf{T}$ Spectral theorem

*Orthogonally* diagonalizable.





## 3 Eigenspaces are orthogonal for symmetric matrices

Eigenvectors (corresponding to distinct eigenvalues) of a symmetric matrix are orthogonal.

Proof.
Consider $\mathbf{A}\boldsymbol{x} = \lambda_1\boldsymbol{x}$ and $\mathbf{A}\boldsymbol{y} = \lambda_2\boldsymbol{y}$ (with $\lambda_1 \neq \lambda_2$). Then

$$\lambda_1\boldsymbol{x}\cdot\boldsymbol{y} \;=\; \mathbf{A}\boldsymbol{x}\cdot\boldsymbol{y} \;=\; \boldsymbol{x}(\mathbf{A}^\mathsf{T})\boldsymbol{y} \;=\; \boldsymbol{x}\mathbf{A}\boldsymbol{y} \;=\; (\boldsymbol{x}\cdot\boldsymbol{y})\lambda_2.$$

Since $\lambda_1 \neq \lambda_2$ necessarily:

$$\lambda_1(\boldsymbol{x}\cdot\boldsymbol{y}) - \lambda_2(\boldsymbol{x}\cdot\boldsymbol{y}) = 0 \quad\Longrightarrow\quad (\lambda_1-\lambda_2)\boldsymbol{x}\cdot\boldsymbol{y} = 0 \quad\Longrightarrow\quad \boldsymbol{x}\cdot\boldsymbol{y} = 0.$$

□





## 4 Quadratic forms

Quadratic form:

$$\boldsymbol{x}\mathbf{A}\boldsymbol{x}; \quad \text{with} \quad \mathbf{A}^\mathsf{T} = \mathbf{A}$$

Since $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\mathsf{T}$ (with $\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{Q}\mathbf{Q}^\mathsf{T} = \mathbf{I}$), then

$$\boldsymbol{x}\mathbf{A}\boldsymbol{x} = \boldsymbol{x}\mathbf{Q}\mathbf{D}\mathbf{Q}^\mathsf{T}\boldsymbol{x} = (\mathbf{Q}^\mathsf{T}\boldsymbol{x})\mathbf{D}(\mathbf{Q}^\mathsf{T}\boldsymbol{x}) \text{ (weighted sum of squares)}$$

Positive definite quadratic form:

$$\boldsymbol{x}\mathbf{A}\boldsymbol{x} > 0 \quad \forall\boldsymbol{x} \neq \mathbf{0} \qquad \Longleftrightarrow \qquad \lambda_i > 0, \quad i = 1:n.$$

Then we also say that $\mathbf{A}$ is positive definite.





## 5 Positive definite matrices

**Meaning:**

$$\boldsymbol{x}\mathbf{A}\boldsymbol{x} > 0 \qquad (\text{except for } \boldsymbol{x} = \mathbf{0})$$

**Some properties**

Consider a positive definite matrix $\mathbf{A}$: What about $\mathbf{A}^{-1}$?
$\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{-1} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\mathsf{T}$

Consider two positive definite matrices $\mathbf{A}$ and $\mathbf{B}$: What about $\mathbf{A} + \mathbf{B}$?

Thus the answer is. . .

## 6 Matrix Product $\mathbf{A}^\mathsf{T}\mathbf{A}$

Let $\underset{m\times n}{\mathbf{A}}$ be a rectangular matrix. Is $\mathbf{A}^\mathsf{T}\mathbf{A}$ positive definite?

$$\boldsymbol{x}(\mathbf{A}^\mathsf{T}\mathbf{A})\boldsymbol{x} =$$

It can only be $0$ when $\mathbf{A}\boldsymbol{x}$ is $\mathbf{0}$

How can we guarantee that $\mathbf{A}\boldsymbol{x} \neq \mathbf{0}$ when $\boldsymbol{x} \neq \mathbf{0}$?





## 7 Symmetric matrices: Signs of eigenvalues

Are all $\lambda_i$ positive? Are they negative?

Computing eigenvalues of $\underset{5\times 5}{\mathbf{A}}$ is impossible in general! (5th degree polynomial)

**Good news:** The signs of the pivots of echelon form are the same as the signs of the eigenvalues $\lambda_i$
(If we do not change the sign of the determinant with *Type II* elementary transformations)

Number of positive pivots = Number of positive eigenvalues





## 8 Positive definite symmetric matrices

- All eigenvalues are:
- All pivots are:

$$\begin{bmatrix} 5 & 2 \\ 2 & 3 \end{bmatrix}$$

**Pivots:**

What is the sign of each eigenvalue?

$$\lambda^2 - 8\lambda + 11 = 0 \rightarrow \qquad \lambda = 4 \pm \sqrt{5} > 0$$





**Summary** (for symmetric matrices):

1. Symmetric matrices have *real eigenvalues* and *perpendicular eigenvectors* can be chosen
2. $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\mathsf{T}$ where $\mathbf{Q}$ is orthogonal
3. $\mathbf{A}$ is symmetric if and only if it is *orthogonally* diagonalizable
4. The signs of the pivots in the echelon form are the same as the signs of the eigenvalues $\lambda_i$ (only if we do not change the sign of the determinant with *Type II* elementary transformations)

## Questions of the Lecture 18

(L-18) Question 1. Write the matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ in the form $\mathbf{Q}\mathbf{D}\mathbf{Q}^\intercal$ of the spectral theorem:

(a) $\mathbf{A} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$

(b) $\mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(c) $\mathbf{C} = \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$

(Strang, 2006, exercise 11 from section 5.5.)

(L-18) Question 2. Find the eigenvalues and the unit eigenvectors (of unit length) of

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

(Strang, 2003, exercise 3 from section 6.4.)

(L-18) Question 3. Find an orthonormal matrix $\mathbf{Q}$ that diagonalizes this symmetric matrix:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & -2 \\ 2 & -2 & 0 \end{bmatrix}$$





(Strang, 2003, exercise 5 from section 6.4.)

(L-18) Question 4. Suppose that $\mathbf{A}$ is a symmetric $3 \times 3$ matrix with eigenvalues 0, 1, and 2.

(a) What properties can be guaranteed for the unit eigenvectors $\boldsymbol{u}$, $\boldsymbol{v}$, and $\boldsymbol{w}$ corresponding to the respective eigenvalues 0, 1, and 2?

(b) In terms of $\boldsymbol{u}$, $\boldsymbol{v}$, and $\boldsymbol{w}$, describe the null space $\mathcal{N}(\mathbf{A})$, the left null space $\mathcal{N}(\mathbf{A}^\intercal)$, the row space $\mathcal{C}(\mathbf{A}^\intercal)$, and the column space $\mathcal{C}(\mathbf{A})$.

(c) Find a vector $\boldsymbol{x}$ such that $\mathbf{A}\boldsymbol{x} = \boldsymbol{v} + \boldsymbol{w}$. Is $\boldsymbol{x}$ unique?

(d) What conditions must we impose on $\boldsymbol{b}$ for $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ to have a solution?

(e) If $\boldsymbol{u}$, $\boldsymbol{v}$, and $\boldsymbol{w}$ are the columns of $\mathbf{S}$, and $\boldsymbol{v}$ is orthogonal to $\boldsymbol{w}$; write the matrices $\mathbf{S}^{-1}$ and $\mathbf{S}^{-1}\mathbf{A}\mathbf{S}$.

(Strang, 2006, exercise 13 from section 5.5.)

(L-18) Question 5. Write a key fact about the eigenvalues of each of these types of matrices:

(a) A real symmetric matrix.

(b) A diagonalizable matrix such that $\mathbf{A}^n \to \mathbf{0}$ as $n \to \infty$.

(c) A non-diagonalizable matrix.

(d) A singular matrix.

(Strang, 2006, exercise 16 from section 5.5.)





(L-18) Question 6. Let

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}; \qquad \mathbf{B} = \frac{1}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

(a) Find the eigenvalues of $\mathbf{A}$ (remember that $i^2 = -1$).

(b) Find the eigenvalues of $\mathbf{B}$ (in this case, it may be easier to first find the eigenvectors, and then deduce the eigenvalues).

(Strang, 2006, exercise 14 from section 5.5.)

(L-18) Question 7. If $\mathbf{A}^3 = \mathbf{0}$ then the eigenvalues of $\mathbf{A}$ must be _____. Provide an example such that $\mathbf{A} \neq \mathbf{0}$. Furthermore, if $\mathbf{A}$ is also symmetric, prove that $\mathbf{A}^3$ is necessarily $\mathbf{0}$.





(L-18) Question 8. Consider the matrix

$$\mathbf{A} = \begin{bmatrix} a & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}.$$

(a) Prove that $\mathbf{A}$ is not diagonalizable when $a = 3$.

(b) Is $\mathbf{A}$ diagonalizable when $a = 2$? (explain). If it is diagonalizable, find an eigenvalue diagonal matrix $\mathbf{D}$ and an eigenvector matrix $\mathbf{S}$ such that $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$.

(c) Is $\mathbf{A}^\intercal\mathbf{A}$ diagonalizable for any value of $a$? Is it possible to find a full set of orthonormal eigenvectors of $\mathbf{A}^\intercal\mathbf{A}$?

(d) Find all values of $a$ for which $\mathbf{A}$ is invertible and diagonalizable.

(L-18) Question 9. Sea la matriz

$$\mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix};$$

(a) Exprese $\mathbf{B}$ en la forma $\mathbf{B} = \mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\intercal$ del teorema espectral.

(b) ¿Es $\mathbf{B}$ diagonalizable? Si no lo es, diga las razones; y en caso contrario genere una matriz $\mathbf{S}$ que diagonalice a $\mathbf{B}$.

(L-18) Question 10. Suppose the vectors $\boldsymbol{q}_1$, $\boldsymbol{q}_2$, $\boldsymbol{q}_3$ form an orthonormal basis for $\mathbb{R}^3$ and the matrix $\mathbf{A}$ satisfies $\mathbf{A}\boldsymbol{q}_1 = (1, 0, 0, )$, $\mathbf{A}\boldsymbol{q}_2 = (0, 1, 0, )$, and $\mathbf{A}\boldsymbol{q}_3 = (0, 0, 1, )$.

(a) ($0.5^{\text{pts}}$) Write the matrix $\mathbf{A}$ explicitly in terms of the vectors $\boldsymbol{q}_1$, $\boldsymbol{q}_2$, $\boldsymbol{q}_3$.
(b) ($1^{\text{pts}}$) Write down all possibilities for $\det \mathbf{A}$.
(c) Which of the following statements are correct: The eigenvalues of $\mathbf{A}$ must...

- be real numbers.
- be positive real numbers.
- be imaginary numbers.
- have absolute value $|\lambda| = 1$.





## 1 Highlights of Lesson 19

### Highlights of Lesson 19

- Positive and Negative (semi)definite matrices
- Completing the square
- Diagonalization by congruence





## 2 Quadratic forms

- Positive definite: $\forall \boldsymbol{x} \neq \mathbf{0} \;\Rightarrow\; \boldsymbol{x}\mathbf{A}\boldsymbol{x} > 0.$
- Positive semi-definite: $\forall \boldsymbol{x} \neq \mathbf{0} \;\Rightarrow\; \boldsymbol{x}\mathbf{A}\boldsymbol{x} \geq 0.$
- Negative definite: $\forall \boldsymbol{x} \neq \mathbf{0} \;\Rightarrow\; \boldsymbol{x}\mathbf{A}\boldsymbol{x} < 0.$
- Negative semi-definite: $\forall \boldsymbol{x} \neq \mathbf{0} \;\Rightarrow\; \boldsymbol{x}\mathbf{A}\boldsymbol{x} \leq 0.$
- Indefinite: neither positive semi-definite nor negative semi-definite.





### Example

What number do I have to put for the matrix $\mathbf{A}$ to be singular?

$$\mathbf{A} = \begin{bmatrix} 2 & 6 \\ 6 & \end{bmatrix}$$

- Eigenvalues:
- Leading principal minors:
- For the quadratic form

$$q_{\mathbf{A}}(\boldsymbol{x}) = \boldsymbol{x}\mathbf{A}\boldsymbol{x} = \begin{pmatrix} x, & y, \end{pmatrix} \begin{bmatrix} 2 & 6 \\ 6 & \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2x^2 + 12xy + \quad y^2$$

Is there a $\boldsymbol{x} \neq \mathbf{0}$ such that $\boldsymbol{x}\mathbf{A}\boldsymbol{x} = 0$?

## Example

If $\mathbf{A} = \begin{bmatrix} 2 & 6 \\ 6 & 7 \end{bmatrix}$ then $\begin{pmatrix} x, & y, \end{pmatrix} \begin{bmatrix} 2 & 6 \\ 6 & 7 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2x^2 + 12xy + \,?\, y^2$

- Are there numbers $x$ and $y$ that make $\boldsymbol{x}\mathbf{A}\boldsymbol{x}$ negative?
- Does the function go through the origin?
- When $y = 0$ and $x = 1$, is it positive? (and when $x = -1$?)
- When $x = 0$ and $y = 1$, is it positive? (and when $y = -1$?)
- Is it always positive?

$(0, 0, )$ **saddle point**: minimum in some directions, maximum in others.

$$\lambda_1 = -2, \quad \begin{pmatrix} -6 \\ 4 \end{pmatrix}; \qquad \lambda_2 = 11, \quad \begin{pmatrix} 6 \\ 9 \end{pmatrix}$$





## Example

If $\mathbf{A} = \begin{bmatrix} 2 & 6 \\ 6 & 20 \end{bmatrix}$ then $\begin{pmatrix} x, & y, \end{pmatrix} \begin{bmatrix} 2 & 6 \\ 6 & 20 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2x^2 + 12xy + 20y^2$

Positive definite.

Tests to check if $\mathbf{A}$ is positive definite

- Are the leading principal minors positive?
- Are the eigenvalues positive?

$$q_{\mathbf{A}}(\boldsymbol{x}) = \boldsymbol{x}\mathbf{A}\boldsymbol{x} > 0 \qquad \text{for all } \boldsymbol{x} \neq \mathbf{0}$$





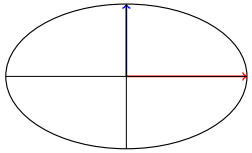
## 3 Completing the squares

If we could express $q(\boldsymbol{x})$ as a sum of squares, we would know whether $q(\boldsymbol{x})$ is positive definite.

Let's complete the square!

- $q(x, y) = 2x^2 + 12xy + 20y^2 \;=\; 2(x + \,?\, y)^2 + \,?$
- $q(x, y) = 2x^2 + 12xy + 7y^2$
- $q(x, y) = 2x^2 + 12xy + 18y^2$
- $q(x, y) = 2x^2 + 12xy + 200y^2$ (graph)

If it is positive definite: $q(x, y) = a$; $a > 0$: ellipse



is

$q(x, y, z, w, t) = 2t^2 - 2tx - 2tz + w^2 - 2wy + 2x^2 - 2xy + 2y^2 + z^2$

positive definite? ☹☹☹☹☹!!!???





## 4 Congruent matrices

**A** and **C** are congruent if there exists an invertible **B** such that

$\boxed{\mathbf{C} = \mathbf{B}^\intercal\mathbf{A}\mathbf{B}}$

Diagonalization by congruence

For every **A** (symmetric) there exists $\mathbf{B} = \mathbf{I}_{\tau_1 \cdots \tau_k}$ (invertible) such that

$$\mathbf{D} = \mathbf{B}^\intercal\mathbf{A}\mathbf{B} \qquad \text{is diagonal} \qquad (\mathbf{B}^\intercal = {}_{\tau_k \cdots \tau_1}\mathbf{I})$$

Spectral Theorem: ¡Diagonalization by similarity and congruence!

$$\mathbf{D} = \mathbf{Q}^{-1}\mathbf{A}\mathbf{Q} = \mathbf{Q}^\intercal\mathbf{A}\mathbf{Q}.$$

Hence, every quadratic form can be expressed as a sum of squares.

$$\boldsymbol{x}\mathbf{A}\boldsymbol{x} \;=\; \boldsymbol{x}\left(\mathbf{B}^{-1}\right)^\intercal\mathbf{D}\mathbf{B}^{-1}\boldsymbol{x} \;=\; \boldsymbol{y}\mathbf{D}\boldsymbol{y}; \qquad \text{where} \qquad \boldsymbol{y} = \mathbf{B}^{-1}\boldsymbol{x}.$$

## 5 Completing the squares

$$2x^2 + 12xy + 20y^2$$

$$\begin{bmatrix} 2 & 6 \\ 6 & 20 \end{bmatrix} \xrightarrow{\substack{\tau \\ [(-3)\mathbf{1}+\mathbf{2}]}} \begin{bmatrix} 2 & 0 \\ 6 & 2 \end{bmatrix} \xrightarrow[\substack{\tau \\ [(-3)\mathbf{1}+\mathbf{2}]}]{} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix};$$

therefore, we have that:

$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \mathbf{D} = \mathbf{E}^\intercal \mathbf{A} \mathbf{E} = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 6 \\ 6 & 20 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix}$$

hence $\mathbf{A} = (\mathbf{E}^\intercal)^{-1} \mathbf{D} \mathbf{E}^{-1}$ and so

$$\boldsymbol{x}\mathbf{A}\boldsymbol{x} = (x, \quad y,) \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \left(\boldsymbol{x}\,(\mathbf{E}^{-1})^\intercal\right) \mathbf{D} \left(\mathbf{E}^{-1}\boldsymbol{x}\right)$$

$$= \left((x+3y), \quad y,\right) \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{pmatrix} (x+3y) \\ y \end{pmatrix} = 2(x+3y)^2 + 2y^2$$





## 6 Example 3 by 3

Is $\mathbf{A} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ positive definite?

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \xrightarrow[\substack{\tau \\ [(\frac{1}{2})\mathbf{1}+\mathbf{2}]}]{\substack{\tau \\ [(\frac{1}{2})\mathbf{1}+\mathbf{2}]}} \begin{bmatrix} 2 & 0 & 0 \\ 0 & \frac{3}{2} & -1 \\ 0 & -1 & 2 \end{bmatrix} \xrightarrow[\substack{\tau \\ [(\frac{2}{3})\mathbf{2}+\mathbf{3}]}]{\substack{\tau \\ [(\frac{2}{3})\mathbf{2}+\mathbf{3}]}} \begin{bmatrix} 2 & 0 & 0 \\ 0 & \frac{3}{2} & 0 \\ 0 & 0 & \frac{4}{3} \end{bmatrix}$$

$$\boldsymbol{x}\mathbf{A}\boldsymbol{x} = 2x^2 + 2y^2 + 2z^2 - 2xy - 2yz > 0$$



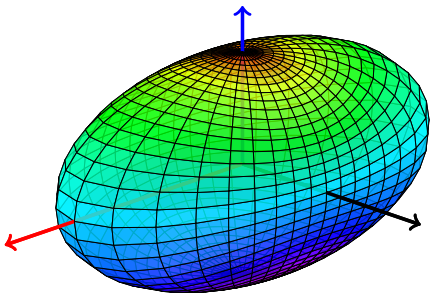
$\boldsymbol{x}\mathbf{A}\boldsymbol{x} = 1$ : (ellipsoid) Axes are in the direction of the eigenvectors

$\mathbf{A} = \mathbf{Q}^\intercal \boldsymbol{\lambda} \mathbf{Q}$



## 7 Positive definite matrices and ellipsoids: example 3 by 3

- The region $(\boldsymbol{x}\mathbf{A}\boldsymbol{x} = a)$ is an (ellipsoid).
- The columns of $\mathbf{Q}$ (eigenvectors of $\mathbf{A}$) are the principal axes.
- Lengths of the axes are determined by the eigenvalues.







## 8 Another example 3 by 3

Is $\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ positive definite?

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \xrightarrow[\substack{\tau \\ [(1)\mathbf{3}+\mathbf{1}]}]{\substack{\tau \\ [(1)\mathbf{3}+\mathbf{1}]}} \begin{bmatrix} 2 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \xrightarrow[\substack{\tau \\ [(-\frac{1}{2})\mathbf{1}+\mathbf{3}]}]{\substack{\tau \\ [(-\frac{1}{2})\mathbf{1}+\mathbf{3}]}} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} \end{bmatrix} \xrightarrow[\substack{\tau \\ [\mathbf{2}\rightleftharpoons\mathbf{3}]}]{\substack{\tau \\ [\mathbf{2}\rightleftharpoons\mathbf{3}]}} \begin{bmatrix} 2 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Indefinite matrix.

## 9 "Classification" of quadratic forms

$$\boxed{\boldsymbol{x}\mathbf{A}\boldsymbol{x} \gtreqless 0; \quad \text{for all } \boldsymbol{x} \neq \mathbf{0}}$$

### Methods

Check the signs of

1. Elem. diag.: $\mathbf{D} = \mathbf{B}^{\intercal}\mathbf{A}\mathbf{B}$ (Diagonalization by congruence) ☺
2. Computing eigenvalues: (Roots of a polynomial) ☹
3. Leading principal minors: (Sylvester's criterion) ☹

### Law of inertia

the number of positive, negative, and zero entries on the diagonal of $\mathbf{D}$ is an invariant of $\mathbf{A}$, i.e., it does not depend on $\mathbf{B}$

(Orthogonal diagonalization $\mathbf{D} = \mathbf{Q}^{\intercal}\mathbf{A}\mathbf{Q}$ is a special case)





## Questions of the Lecture 19

(L-19) Question 1. Decide whether the following matrices are positive definite, and write out the corresponding quadratic forms $f = \boldsymbol{x}\mathbf{A}\boldsymbol{x}$ :

(a) $\begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$

(d) $\begin{bmatrix} -1 & 2 \\ 2 & -8 \end{bmatrix}$

(e) The determinant in (b) is zero; along what line does $f(x, y) = 0$?

(Strang, 2006, exercise 2 from section 6.1.)

(L-19) Question 2. What is the quadratic form $f = ax^2 + 2bxy + cy^2$ for each of these matrices? Complete the square to write $f$ as a sum of one or two squares $d_1(\quad)^2 + d_2(\quad)^2$.

(a) $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 9 \end{bmatrix}$

(b) $\mathbf{B} = \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}$

(Strang, 2006, exercise 15 from section 6.1.)





(L-19) Question 3. Which of the following matrices has two positive eigenvalues? Test $a > 0$ and $ac > b^2$, do not compute the eigenvalues. $\boldsymbol{x}\mathbf{A}\boldsymbol{x} < 0$.

(a) $\mathbf{A} = \begin{bmatrix} 5 & 6 \\ 6 & 7 \end{bmatrix}$

(b) $\mathbf{B} = \begin{bmatrix} -1 & -2 \\ -2 & -5 \end{bmatrix}$

(c) $\mathbf{C} = \begin{bmatrix} 1 & 10 \\ 10 & 100 \end{bmatrix}$

(d) $\mathbf{D} = \begin{bmatrix} 1 & 10 \\ 10 & 101 \end{bmatrix}$

(Strang, 2006, exercise 14 from section 6.1.)

(L-19) Question 4. Show that $f(x, y) = x^2 + 4xy + 3y^2$ does not have a minimum at $(0, 0)$ even though it has positive coefficients. Write $f(x, y)$ as a difference of squares and find a point $(x, y)$ where $f(x, y)$ is negative.
(Strang, 2006, exercise 16 from section 6.1.)

(L-19) Question 5. Show from the eigenvalues that if $\mathbf{A}$ is positive definite, so is $\mathbf{A}^2$ and so is$\mathbf{A}^{-1}$.
(Strang, 2006, exercise 4 from section 6.2.)





(L-19) Question 6. Consider the following quadratic forms

$$\begin{aligned} q_1(x, y, z) &= x^2 + 4y^2 + 5z^2 - 4xy. \\ q_2(x, y, z) &= -x^2 + 4y^2 + z^2 + 2xy - 2axz. \end{aligned}$$

(a) Show that $q_1(x, y, z)$ is positive semi-definite.
(b) Find, if it is possible, any value of $a$ such that $q_2(x, y, z)$ is negative definite.

(L-19) Question 7. Decide whether the following matrices are positive definite or not.

(a) $\mathbf{A} = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$

(b) $\mathbf{B} = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}$

(c) $\mathbf{C} = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}^2$

(Strang, 2006, exercise 2 from section 6.2.)

(L-19) Question 8. Classify the quadratic form (i.e., positive definite, negative, semi-definite, indefinite, etc.)

$$q(x, y, z) = x^2 + 6xy + y^2 + az^2;$$

in terms of the parameter $a$.

(L-19) Question 9. If $\mathbf{A} = \left[\begin{smallmatrix} a & b \\ b & d \end{smallmatrix}\right]$ is positive definite, prove that $\mathbf{A}^{-1}$ is positive definite.
(Strang, 2006, exercise 8 from section 6.1.)

(L-19) Question 10. If a symmetric matrix of 2 by 2 satisfies $a > 0$, and $ac > b^2$, prove that its eigenvalues are real and positive (positive definite). Use the characteristic equation and the fact that the product of the eigenvalues equals the determinant.
(Strang, 2006, exercise 3 from section 6.1.)





(L-19) Question 11. Decide whether the following matrices are positive definite, negative definite, semi-definite, or indefinite.

(a) $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 4 \\ 3 & 4 & 9 \end{bmatrix}$
(b) $\mathbf{B} = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 2 & 6 & -2 & 0 \\ 0 & -2 & 5 & -2 \\ 0 & 0 & -2 & 3 \end{bmatrix}$
(c) $\mathbf{C} = -\mathbf{B}$
(d) $\mathbf{D} = \mathbf{A}^{-1}$

(L-19) Question 12. A positively defined matrix cannot have a zero (or worse; a negative number) on its main diagonal. Prove that this matrix does not satisfy. $\boldsymbol{x}\mathbf{A}\boldsymbol{x} > 0$, for all $\boldsymbol{x} \neq \mathbf{0}$:

$$\begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix} \begin{bmatrix} 4 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 5 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \quad isnotpositivewhen \quad \begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix} = \begin{pmatrix} \quad & \quad & \quad \end{pmatrix}$$

(Strang, 2006, exercise 21 from section 6.2.)





(L-19) Question 13. Demonstrate that if $\mathbf{A}$ and $\mathbf{B}$ are positive definite then $\mathbf{A} + \mathbf{B}$ is also positive definite. For this demonstration, pivots and eigenvalues are not convenient. It is better to use $\boldsymbol{x}(\mathbf{A} + \mathbf{B})\boldsymbol{x} > 0$
(Strang, 2006, exercise 5 from section 6.2.)

(L-19) Question 14. Find the $\dot{\mathbf{L}}\mathbf{D}\dot{\mathbf{L}}^{\mathsf{T}}$ factorization for the following symmetric matrices.

(a)

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & 20 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{bmatrix}$$

(L-19) Question 15. The quadratic form $f(x, y) = 3(x + 2y)^2 + 4y^2$ is positive definite. Find the matrix $\mathbf{A}$, factor it into $\mathbf{L}\mathbf{D}\mathbf{L}^{\mathsf{T}}$, and relate the elements in $\mathbf{D}$ and $\mathbf{L}$ with 3, 2 and 4 in $f$.
(Strang, 2006, exercise 9 from section 6.1.)





(L-19) Question 16. Consider the following matrices

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & a & 0 \\ 0 & 0 & a & a \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(a) $(0.5^{\text{pts}})$ Compute the eigenvalues of $\mathbf{A}$.
(b) $(0.5^{\text{pts}})$ Prove that when $a = 2$ the matrix $\mathbf{A}$ is not diagonalizable.
(c) $(1^{\text{pts}})$ For matrix $\mathbf{B}$, find a diagonal matrix $\mathbf{D}$ and an orthonormal matrix $\mathbf{P}$ such that $\mathbf{B} = \mathbf{P}\mathbf{D}\mathbf{P}^{\mathsf{T}}$.
(d) $(0.5^{\text{pts}})$ Find the polynomial expression of the quadratic form associated with the matrix $\mathbf{B}$, and prove that it is positive definite.

(L-19) Question 17. Given the matrix $\mathbf{A} = \left(\begin{smallmatrix} a & 3/5 \\ b & 4/5 \end{smallmatrix}\right)$, compute the values (if they exist) of $a$ and $b$ such that

(a) $(0.5^{\text{pts}})$ The matrix $\mathbf{A}$ is orthonormal.
(b) $(0.5^{\text{pts}})$ The columns of $\mathbf{A}$ are linearly independent.
(c) $(0.5^{\text{pts}})$ $\lambda = 0$ is an eigenvalue of $\mathbf{A}$.
(d) $(0.5^{\text{pts}})$ $\mathbf{A}$ is symmetric and negative definite.

(L-19) Question 18.

(a) Obtain the matrix $\mathbf{Q}$ associated with the quadratic form $q(x, y, z) = x^2 + 2xy + ay^2 + 8z^2$ and classify the matrix $\mathbf{Q}$ (that is, indicate in which cases the matrix is indefinite, definite, or semidefinite, either positive or negative) for the case when $a$ is equal to one $(a = 1)$.
(b) Classify the matrix $\mathbf{Q}$ when $a \neq 1$.





## Questions of the Optional Lecture 2

(L-Opt-2) Question 1. Consider the following matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & a & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

(a) $(0.5^{\text{pts}})$ Show that $\mathbf{A}$ is invertible if and only if $a \neq 0$.
(b) $(0.5^{\text{pts}})$ Is $\mathbf{A}$ positive definite when $a = 1$? Justify your answer.
(c) $(1^{\text{pts}})$ Compute $\mathbf{A}^{-1}$ when $a = 2$.
(d) $(0.5^{\text{pts}})$ How many variables can be chosen as exogenous (or free) variables in the system $\mathbf{A}\boldsymbol{x} = \mathbf{0}$ when $a = 0$? Which ones?

(L-Opt-2) Question 2. True or false (to receive full credit you must justify your answers clearly and concisely)

(a) If $\mathbf{A}$ is symmetric, then so is $\mathbf{A}^2$.
(b) If $\mathbf{A}^2 = \mathbf{A}$ then $(\mathbf{I} - \mathbf{A})^2 = (\mathbf{I} - \mathbf{A})$ where $\mathbf{I}$ is the identity matrix.
(c) If $\lambda = 0$ is an eigenvalue of the squared matrix $\mathbf{A}$, then the linear system $\mathbf{A}\boldsymbol{x} = \mathbf{0}$ is is always solvable and has only one solution.
(d) If $\lambda = 0$ is an eigenvalue of the squared matrix $\mathbf{A}$, then the linear system $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ may be inconsistent.
(e) If a matrix is orthogonal (columns are orthogonal and of unit length), then its inverse is also orthogonal.





(f) If 1 is the only eigenvalue of a $2 \times 2$ matrix $\mathbf{A}$, then $\mathbf{A}$ must be the identity matrix $\mathbf{I}$.

(L-Opt-2) Question 3. fill in the blanks, or respond **T**rue/**F**alse.

(a) Any generating system of a vector space contains a basis of the space (T/F)

_____.
(b) Every vector space is finite dimensional (T/F) _____.
(c) That $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n$ are linearly independent means that

_____.

(d) The set that contains only the vector $\mathbf{0}$ is a linearly independent set. (T/F)

_____.
(e) A square matrix of order $n$ by $n$ is diagonalizable when:

_____.

(f) If $\boldsymbol{u} = (1, 2, -1, 1)$, then $||\boldsymbol{u}|| =$ _____.
(g) If $\boldsymbol{u} = (1, 2, -1, 1)$ and $\boldsymbol{v} = (-2, 1, 0, 0)$, then $\boldsymbol{u} \cdot \boldsymbol{v} =$ _____.





(L-Opt-2) Question 4. In the following questions, $\mathbf{A}$ and $\mathbf{B}$ are $n \times n$ matrices. Indicate whether the following statements are true or false (include a brief explanation or a counterexample that justifies your answer):

(a) If $\mathbf{A}$ is not zero, then $\det(\mathbf{A}) \neq 0$.
(b) If $\det(\mathbf{AB}) \neq 0$, then $\mathbf{A}$ is invertible.
(c) If I swap the first two rows of $\mathbf{A}$, its eigenvalues change.
(d) If $\mathbf{A}$ is real and symmetric, then its eigenvalues are real **(no justification is needed here)**.
(e) If the reduced echelon form of $(\mathbf{A} - 5\mathbf{I})$ is the identity matrix, then 5 is not an eigenvalue of $\mathbf{A}$.
(f) Let $\boldsymbol{b}$ a column vector in $\mathbb{R}^n$. If the system $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ has no solution, then $\det(\mathbf{A}) \neq 0$
(g) Let $\mathbf{C}$ of size $3 \times 5$. The rank of $\mathbf{C}$ can be 4.
(h) Let $\mathbf{C}$ of size $n \times m$, and $\boldsymbol{b}$ a column vector in $\mathbb{R}^n$. If $\mathbf{C}\boldsymbol{x} = \boldsymbol{b}$ has no solution, then $\operatorname{rg}(\mathbf{C}) < n$.
(i) Every diagonalizable matrix is invertible.
(j) If $\mathbf{A}$ is invertible, then su forma reducida de echelon is the identity matrix.

(L-Opt-2) Question 5. Sean Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 4 \\ 0 & 0 & 5 \end{bmatrix}; \quad \mathbf{B} = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 1 & 3 \\ 0 & 0 & 2 \end{bmatrix}; \quad \mathbf{C} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 0 & 4 \\ 0 & 0 & 6 \end{bmatrix}; \quad \mathbf{D} = \begin{bmatrix} 2 & 3 & 0 \\ 3 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The eigenvalues of $\mathbf{B}$ are 0 and 2. Use this information to answer the following questions. For each matrix, provide an explanation. There may be more than one matrix that satisfies the condition:

(a) Which matrices are invertible?
(b) Which matrices have a repeated eigenvalue?
(c) Which matrices have a rank less than three?
(d) Which matrices are diagonalizable?
(e) For which diagonalizable matrices can we find three mutually orthogonal eigenvectors?

(L-Opt-2) Question 6. Consider the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

(a) Find the eigenvalues and eigenvectors of $\mathbf{A}$.
(b) Is $\mathbf{A}$ diagonalizable?





(a) What are its column space $\mathcal{C}(\mathbf{A})$ and null space $\mathcal{N}(\mathbf{A})$? (do not answer with the definition, indicate which set of vectors composes each space).
(b) Suppose that $\mathbf{A}$ can be factored as $\mathbf{A} = \mathbf{LU}$:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 7 & 3 & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

Describe the first elimination step in reducing $\mathbf{A}$ to $\mathbf{U}$. How do you know that $\mathbf{U}$ is also an invertible matrix? What is the value of the determinant of $\mathbf{A}$?
(c) Find a particular $3 \times 3$ invertible matrix $\mathbf{A}$ that cannot be factored in the form $\mathbf{LU}$ (without first permuting the rows). What factorization is still possible in your example? (it is not necessary to perform the factorization). How do you know that your matrix $\mathbf{A}$ is invertible?

Strang, G. (2003). *Introduction to Linear Algebra*. Wellesley-Cambridge Press, Wellesley, Massachusetts. USA, third ed. ISBN 0-9614088-9-8.

Strang, G. (2006). *Linear algebra and its applications*. Thomson Learning, Inc., fourth ed. ISBN 0-03-010567-6.





(c) Is it possible to find a matrix $\mathbf{P}$ such that $\mathbf{A} = \mathbf{PDP}^\intercal$, where $\mathbf{D}$ is a diagonal matrix?
(d) Compute $|\mathbf{A}^{-1}|$.

(L-Opt-2) Question 7. Consider a $3 \times 3$ matrix $\mathbf{A}$ with eigenvalues $\lambda_1 = 1$, $\lambda_2 = 2$, and $\lambda_3 = -1$; and let $\boldsymbol{v}_1 = (1, 0, 1)^\intercal$ and $\boldsymbol{v}_2 = (1, 1, 1)^\intercal$ be the corresponding eigenvectors to $\lambda_1$ and $\lambda_2$.

(a) Is $\mathbf{A}$ diagonalizable?
(b) Could $\boldsymbol{v}_3 = (-1, 0, -1)^\intercal$ be an eigenvector associated with the eigenvalue $\lambda_3 = -1$?
(c) Compute $\mathbf{A}(\boldsymbol{v}_1 - \boldsymbol{v}_2)$.

(L-Opt-2) Question 8.

(a) ($0.5^{\text{pts}}$) Find a homogeneous system $\mathbf{A}\boldsymbol{x} = \mathbf{0}$ whose solution set is

$$\left\{ \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} \in \mathbb{R}^4 \;\middle|\; \exists \alpha, \beta, \gamma \in \mathbb{R} \quad \text{such that} \quad \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} + \gamma \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \right\}$$

(b) ($0.5^{\text{pts}}$) If the characteristic polynomial of the matrix $\mathbf{A}$ is $p(\lambda) = \lambda^5 + 3\lambda^4 - 24\lambda^3 + 28\lambda^2 - 3\lambda + 10$, find the rank of $\mathbf{A}$.

(L-Opt-2) Question 9. Suppose a square and *invertible* matrix $\underset{n \times n}{\mathbf{A}}$.